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Investigating Benefits of Blockchain Technology on Small Companies

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I. Introduction

a. Background

Blockchain technology became accessible after Satoshi Nakamoto published Bitcoin cryptocurrency in 2008 (Markunas et al., 2019). Since then, the technology has grown and now offers more than just cryptocurrency transfers (Salman et al., 2019). The purpose of this project is to analyse Satoshi Nakamoto's invention and further blockchain developments to find benefits that could be used by small companies. While the first generation of this technology was designed for anonymous money transactions, since then, it has developed and expanded, creating more practical uses for day-to-day businesses (Salman et al., 2019).

Blockchain works as a decentralized database, also called a ledger, which is distributed on the peer-to-peer network, making all data on the blockchain accessible to everyone (Ali Syed et al., 2019). On a public blockchain, anyone can access the data without providing personal details. All transactions on the network are validated through proof-of-work (POW) which will be explained in section II-d. Bitcoin is an example of a public ledger that also has a Blockchain Improvement Proposal available on GitHub.com where anyone in the community can have an impact on future bitcoin updates (Anjum et al., 2017). This can only be found on a public blockchains. Private blockchains are developed by companies that have most, if not

all control over the updates. Hyperledger, developed and released in 2015 by Linux Foundation, is another great example of an open-source blockchain which was created for building robust applications and hardware systems for industry-specific business transactions. It is a private blockchain that requires permission to be accessed (Anjum et al., 2017; Linux Foundation, 2016). Several corporations are already investigating and contributing to the technology development, such as IBM, R3, Accenture and Cisco (Linux Foundation, 2016).

In 2009, Bitcoin started with 1 megabyte of data which grew in 2020 to over 302 gigabytes according to the statistics on Statista.com by Roynor de Best from 2020 (de Best, 2020). With the growth of speed, Bitcoin is predicted to grow up to 3.2tbtes in 2022 (Xu & Huang, 2020). The statistics are very similar to Ethereum, the second most popular public blockchain. This just shows the expansion and progress of technology over the past 12 years due to the involvement of the community and corporations. Many companies see the potential of technology with benefits such as a decentralized nature, anonymity, auditability, or transparency.

The third generation of the technology introduced smart contracts which are used to create strict agreements, also referred to as instructions (Salman et al., 2019). Smart contracts are self-executed scripts, triggered if programmed specifications are met. Those

can be money transfers, messages, or any programmed event on the blockchain. Research shows that smart contracts can be used for complex tasks and overseas transactions between two parties (Valtanen et al., 2019).

b. Project Motivation

Twelve years later, corporations are finding new uses for the technology with the success of developing new blockchains, yet there are no solutions to common problems which small companies face during their daily activities. Blockchain started in the financial sector as mentioned before, but it does not have to be limited to this (Gamache, 2020). Research by Business Insider (2020) and Forbes (2020) proves that blockchains can be implemented into the supply chain, human resources, and asset management; those areas will be the main focus of this project as they show the most potential to be useful for small companies (Business Insider, 2020; Castillo et al., 2020).

c. Purpose

This project aims to find how blockchains can help small businesses financially or technologically. Research shows that businesses can struggle with managerial aspects in their systems; they cannot keep up with their workflow, financial planning, and human resources (Kolowich-Cox, 2020). Newly established companies are usually not supported by big investors and need

alternative ideas to expand without spending all of their budget.

The objective to analyse the technology and how big corporations use blockchain which will show how small businesses can also benefit from the techniques of application. Some small businesses have already introduced blockchains in their systems; analysing them will help foresee the financial benefits and their usefulness on a wider scale, looking at different business sectors, identifying their pros and cons.

d. Assumptions/Limitations

There has been a lot of interest in the media and the business world around blockchain technology but to this day, no company used it as the prime function to revolutionise their business model (Business Insider, 2020; Vega & Swift, 2020). Research shows that big companies invest in technology and aim to profit from the innovation, however, do not focus just on blockchains. Section II and III will outline the results of the analysis and assess how small companies can use the technology.

e. Conclusion

Blockchain has the potential to be revolutionary and applicable to all small companies. By following the footsteps of big corporations, this research will show the best uses of technology, but also the development process and simulation of technology application in one of the analysed business areas.

II. Blockchain Research

a. Blockchain Fundamentals

Blockchain is created out of a chain of data blocks, each containing a header and a body. The header stores information on the previous and current block (hash number), creation of the timestamp, a random number (nonce) and the difficulty of cryptographic operations (Shahzad & Crowcroft, 2019). This data defines the block, making it different to the previous as the timestamp and the hash number would only be used once. The body stores blockchain information, such as files or a message to the recipient. For instance, block's body in bitcoin stores electronic cash in a form of a coin encrypted with a hash algorithm, used for unique identification (Shahnaz & Khalid, 2019). It is important to mention, that miners are a required type of users who are responsible for block validation and system decentralization as they store a copy of the whole blockchain (Yang, 2019; Salman et al., 2019).

b. Public vs Private & Consortium

There are three types of blockchain, which determine who has the access to the data. Open blockchain technology, also known as a public blockchain, is accessible and transparent to everyone on the network (Morkunas et al., 2019). A fully decentralized network is created, as anyone has the ability to join, read, write, and participate, but no one has the power to change the existing data. Bitcoin and Ethereum are two of the most

popular public ledgers (Shahzad & Crowcroft, 2019). However, easy access to information by everyone, might not suit companies that value privacy and deal with personal data. The nature of fully decentralized networks can also be problematic due to a lack of control within business environments. Therefore, this would be the most challenging type of blockchain for small businesses, due to factors such as full decentralization and transparency (Shkoor, 2019). Small companies would have to build a system that does not have to be controlled by them to mitigate this issue.

Closed blockchain (Private) can only be accessed by permission from the blockchain owner; in many cases this type of ledger is developed by one company, limiting access to information and private data (Morkunas et al., 2019; Shkoor, 2019). All transactions are faster than the public blockchain as they consist of fewer nodes and miners. However, anonymity on the network is not fully compromised as the user has to be identified. Theoretically, this could be a good solution for small businesses if introduced correctly, as it is vital to store information securely through the verification process which prevents unwanted individuals from accessing them (Massessi, 2018). Therefore, it is easy to scale up, reduce costs, increase security, reliability and the level of trust, as only authorised individuals to have access to the blockchain (Morkunas et al., 2019).

Consortium blockchain i.e. hybrid is based on a closed network that can only be accessed by validated individuals or groups. Quorum is an example of a consortium blockchain (Ali Syed et al., 2019). The difference between consortium and private blockchain is that private blockchain is governed by one company whereas, the consortium has several company authorities. Having more authority over the ledger allows them to design and govern the protocols within the network (Shahzad & Crowcroft, 2019). The network nodes are predefined, minimising the risk of attacks such as “attack 51%”, which is one of the main reoccurring security risks (Price, 2018). Attack 51% appears when hackers are responsible for more than 50% of validation and creation of new blocks, this issue will be elaborated on in section II-f (Chang et al., 2020). Due to minimum risk of the attack 51%, this type of the blockchain could be the best option for companies as they would control all data transmitted on the network. Unfortunately, this type of investment could cost companies a fortune, which is simply unaffordable for small companies (Viryasitavata & Hoonsoponb, 2019).

c. Centralized vs Decentralized Network

In the centralized network, all transactions have to be authenticated and authorized by the body. A bank is a body that has to approve the transaction before the money arrives in the second bank account. It authorises the sender and recipient's credibility which then allows for

the money to be transferred to the right account (Ali Syed et al., 2019). Financial institutions are not the only examples of centralized systems, as these also include for example Google or Yahoo. To use their email service, a user has to create an account, provide personal information linking every activity on the account to the company's centralized servers (Hebert & Cerbo, 2019). This allows for tracking down who is responsible for the process; it is very practical for detecting fraud or breaking of the law. One of the issues with a centralized system network is that it has a single point of failure which can cause the whole system to shut down, as all the data is stored on the server where the malfunction can destroy the data or lose it forever.

Blockchain on other hand offers a decentralized system that trusts all parties and allows for the transaction to be made without the approval of the intermediaries. This is because the authorisation of all transactions on the blockchain is done in a secure, encrypted manner by the miners making it near to impossible to manipulate the transactions; information stored inside is only accessible for the recipient with a specific key to access it. This process is elaborated on further down the analysis (Ali Syed et al., 2019).

d. Blockchain Security and Mining

Blockchains require miners to validate all transactions, authorising the source of the

blocks and creating them on the blockchain (Kim, 2020). They are self-selected members of the public who use their computing power to mine new blocks and automatically create a full copy of the blockchain on their machines. Individual miners have no impact on the data shared across the network. If the blockchain is private, miners have no authority over it, as it is mainly controlled by the body which created the blockchain (Salman et al., 2019). Mining is a very complex and difficult process, but crucial in making blockchain network secure. It uses several protocols which establish if blocks are going to be allowed on the network, creating a barrier from malicious software, and hacking attempts. The miner who creates the block first will be rewarded with a price which usually is in a form of cryptocurrency coins (Ali et al., 2020). There are seven main factors on which a miner can be established as a winner. Those algorithmic factors are as follows:

- Proof of Work (POW) mining technique requires solving a random and challenging mathematical puzzle that creates new blocks. After authentication by another node, blocks are added to the blockchain and the miner is rewarded. This method requires high computational power as it is required to solve the puzzles. This method is used mainly in Bitcoin and Ethereum (Salmon et al., 2019).
- Proof of Stake (POS) is a more complicated method, as a miner is

chosen depending on the state of his node. If a miner has a big portfolio of rewards, he is more likely to be chosen to create the block, but this process can only happen every 30 days to prevent the same miner with a big portfolio from creating all the blocks, keeping the network decentralized (Xu & Huang, 2020).

- Minimum Block Hash is a very similar method of mining as POS, but the randomisation is not dependent on the node's portfolio, but it is fully based on luck (Salmon et al., 2019).
- Proof of Space (PoSpace) is a mining technique where miners provide device storage space to perform required mining, involving a high storage capability (Salmon et al., 2019).
- Proof of Importance (POI) calculates the significance of an individual node by determining which miner is eligible to perform the calculation by assessing and prioritising all the miners/nodes (Salmon et al., 2019).
- Proof of Burn (POB) is a mining process that verifies the completion of POW and POS, which as a result creates the coins (reward) for the miner who completed the mining process first. It can be also described as a fuel to create new blocks and ensure that the individuals are awarded for their work (Shahzad & Crowcroft, 2019).

- Measure of Trust (MOT) is done by analysing nodes' validity, reliability, and governance. If miners follow the blockchain mining rules and protocols they have a higher chance of being chosen (Ali Syed et al., 2019).
- Practical Byzantine Fault Tolerance (pBFT) provides encryption for messages and protects from node failure and safeguards from employing incorrect blocks to the blockchain (Ali Syed et al., 2019). It is a different mining method to others because validation is predetermined, as the system chooses the miner who will create the next block before the transaction begins. Difficulty and the rate of processing messages do not change, creating less competition between the miners for prizes (Sukhwani et al., 2017). This can be found on Hyperledger Fabrics (Malkhi, 2018).

The mining process analysed above is one of the key features making the blockchain secure and reliable. Cryptography is another. The blocks are kept under cryptographic encryption, also called hashing; it is a process of sending data through encrypted services using SHA-256. It produces strings of characters unreadable for humans which can be translated back to English using a special key that decrypts the code, making a very strong barrier, nearly impossible to break

(Lake, 2019; Shahnaz et al., 2019). Unfortunately, only 'nearly' impossible, as there have been instances where for example approximately \$72 million in crypto currencies have been stolen from Bitfinex in Hong Kong (IBM, 2020). This shows that blockchain security features cannot stop the most skilful hackers from achieving their goal. How they achieved this is unknown according to Clare Baldwin for Reuters (Baldwin, 2016). Shahzad & Crowcroft (2019) explored blockchain security and hashing in greater detail in their academic paper dedicated to electronic voting using blockchains (Shahzad & Crowcroft, 2019). Small companies would have to create a profitable mining system on their blockchain solutions to keep the network decentralized and validated. Nonetheless, this technology has many benefits which small companies could use to their advantage.

e. Benefits of the Blockchain

i. Transparency

All data stored on the blockchain is permanent and transparent to everyone in the network (Bandora et al., 2019). All documents in the network are identical. Any changes to the data have to be approved by all participating miners before they are implemented (Hooper, 2018). This could be beneficial for companies' information such as election votes, health records or business expenses ready for financial analysis which need to be accessed by several bodies in the network. It could be

used by small companies to keep information such as product measurements secure.

ii. Traceability

Blockchain offers verification of the history and location by documenting records of goods such as physical items (Heister et al., 2020). When the item is recorded on the blockchain, it stores the location, time and who currently hold the goods, creating a fully traceable history, which can be used to verify the authenticity and prevent fraud (Hooper, 2018). Small companies could use this for tracking food stocks like has been done by Walmart. The company is tracking their organic products to make sure it comes from the authorised source fresh and in good condition (Stanton et al., 2020, p104).

iii. Security

One of the main attributes that blockchain possess, is built-in security which prevents tampering with data as it is physically impossible due to mining and validations. Transactions made by companies such as the exchange of values would be the safest option in the blockchain (Shahnaz et al., 2019).

iv. Efficiency

The decentralized nature of the network ledger helps with many processes by removing the middleman which takes time and charges unnecessary fees. Continental transfers are done more efficiently, which can be beneficial for companies working between several countries. An example of this could be

purchasing materials from other countries on a supply chain without losing the location and information on the transports (Harting et al., 2020).

v. Immutability

Submitted data on the ledger is distributed across all connected network devices making the information unchangeable as all data is verified by the mining process and cannot be altered, making the blockchain immutable. This feature could be used for promoting the company's services as reliable and trustworthy giving them an advantage over the competition (Palik et al., 2019).

vi. Anonymity

Anonymity is in a way a part of the public ledger. The transaction stays fully anonymous as long as a user profile is not connected to the wallet address. The private ledger is more complicated, as to get access to the network, someone from the group has to invite the user, making the anonymity not entirely true as the login can be traced. However, businesses could use this feature for providing services that require anonymity such as voting platforms (Hebert & Cerbo, 2019).

vii. Consistency

Another benefit of using blockchains is that the mining process distributes information equally, maintaining consistency across the network, making the blockchain marketable and reliable to use for the company's services (Hebert & Cerbo, 2019).

viii. Availability

Everyone on the network can host a full replica of blockchain data, which can be accessed from anywhere, as long as one node of the blockchain network is available (Bandora et al., 2019). Making the system accessible all year could be beneficial for companies and individuals in countries with different time zones but also minimising the risk of system downtime which could be beneficial for a small size company (Palik et al., 2019).

f. Limitations of the Blockchain

Limitations such as scalability, flexibility and governance are just some of the problems which are usually referred to when businesses consider using the technology. The scalability problem comes from a lack of processing speed, due to the complexity of the transactions and the difficulty of encryptions. When using blockchain, fewer parties are involved in the transaction process, eliminating banks and financial services such as Visa. Looking at the blockchain (Bitcoin), the only parties involved in the transaction are the recipient and receiver, however only theoretically this method is faster. Bitcoin technology can only process 7 transactions per second in comparison to Visa which manages 24,000 transactions per second. This clearly shows that transaction processes across Bitcoin can take much longer than standard bank transfers (Anjum, 2017; Chang et al., 2020). Ethereum works better with the processing speed of 20 transactions per

second but still does not match traditional transactions. This issue can make public blockchain unnecessary for businesses that require fast transactions, for example, the real estate sector where house bidding or instant buy is essential and asset exchange has to be finalised immediately. Private blockchain and consortium on other hand, do not face this problem, as the closed network will not experience transactions from the public, speeding up the transactions (Harting et al., 2020).

Decentralization of the blockchain can be the second major problem with the technology. This feature can be a very difficult barrier for corporations, as everything uploaded onto the blockchain will stay there forever and cannot be changed in case of human error. Therefore, the only information submitted should be one that can stay in the log forever, for example, well-written invoices which have been programmed correctly the first-time round (Ahluwalia et al., 2016).

Mining could cause trouble for small businesses too, as it is a very complicated process that requires powerful computers and an army of miners for block validation and system decentralisation. For mining to be profitable from the miners' perspective, it has to create a type of reward which would cover the expenses of the mining and generate profit, as powerful computers need to be constantly supplied with electricity. In the case

of a bitcoin, miners are awarded bitcoin cryptocurrency which then can be exchanged to fiat currencies, earning an algorithmic percentage of the coin value. In February 2021, bitcoin value reached over £40 thousand according to Morningstar for “Currency and Coinbase for Cryptocurrency” provided by Google Finance, making it a profitable investment when mined on a larger scale. Research on bitcoin by Hong from Investopedia (2021) shows that miners’ reward halves every 210,000 blocks mined. In 2009, mining of one block was rewarded with 50 (BTC) bitcoins, in 2012 it was only 25 BTC and currently, in 2020 miners only receive 6.26 BTC (Hong, 2021). This shows that over time, the mining process will be less profitable when aiming for stable cryptocurrencies, assuming it will not be increasing in value.

The “51% attack” is a threat of cyber-attacks which could also be an issue for small companies, and it needs to be explained. This attack happens when organised hackers use the mining process to gain more than half of the computing network. This is given by the protocols Proof-Of-Stake and Proof-Of-Space. These protocols are keeping the network decentralized and validated (Chang et al., 2020). Proof-Of-Stake chooses the miner randomly from all the miners on the blockchain, as by choosing the same miner(s) consistently, there is a higher risk of those miners interfering in the process of creating new blocks. This can happen on a small

blockchain with a small number of miners. By controlling the creation of new blocks, miners could cancel the transaction process by crashing the block, preventing other miners from creating blocks. This could cause fundamental issues in reliability (Frankenfield, 2019). Moreover, by creating all-new blocks, hackers could receive all the cryptocurrency rewards for their mining which can be classified as fraud as they could gain power over the cryptocurrency market value. The third and final issue with 51% attack is that miners could reverse the transaction process, meaning that the coin spent in the transaction could end up in the miner’s possession, causing cryptocurrency double-spending (Yang et al., 2019).

On a public blockchain, individuals have the option to choose if they would like to mine. On permission (private) blockchain, multiple different businesses/individuals are required to participate in mining to make the system decentralized. The main problem is that some businesses prefer to create blocks for transactions themselves, disallowing other members participation (Martino, 2019). This is an issue because the company creating the blocks could try to manipulate the transactions as described in the 51% attack. Due to this, it is important to remember that small businesses would have to find a way to create or establish miners for their system to create new blocks. A company could purchase expensive computers which would be

dedicated just for this purpose, but this could be very costly and defeat the purpose of the decentralized system if only one business is responsible for the mining (Martino, 2019). A solution to this could be creating a blockchain where miners are awarded cryptocurrency, the same way as bitcoin, and build strong value cryptocurrency or use existing private blockchains such as IBM Hyperledger, which does not use traditional mining methods such as POW for validation. Hyperledger fabrics use a specific system where transactions are confirmed by all peers using pBFT algorithms (Malkhi, 2018). Sukhwani et al. went into more technical details on the topic to find out if this could be an alternative mining process (Sukhwani et al., 2017). This is mainly beneficial in consortium blockchains where several businesses are responsible for the mining process, stabilising the decentralization and distributing the costs between each company using the technology (Sheffield, 2018).

III. Blockchain Service Analysis

a. Different uses for Blockchain

In 2019 Deloitte conducted a survey on blockchain uses in the business field (Moreano et al., 2019). The 1500 senior executives and practitioners stated that their companies predict to invest more than \$1 million in the next 12 months as they are afraid to lose competitive advantage if they do not adopt the blockchain technology (De Meijer, 2020). Technologies such as digital currencies and

asset management are at the top of the list for areas that corporations are investigating. The study shows that 57% of the companies use blockchain to benefit financially where only 1% is not sure how they can use the technology (Moreano et al., 2019). This proves that companies, including Deloitte, seek the potential in the technology and are determined to explore further uses of the blockchain. Implementation and regulations of the technology are the biggest threats that companies face by investing in 2019 (Hobbs et al., 2018). The survey on the other hand shows the increasing popularity which could mean that companies are finding solutions to overcome the issues (Moreano et al., 2019).

The release of 'Blockchain 50' by Forbes Magazine summarised the biggest companies which use blockchain in 2019 (Vega & Swift, 2020). Search giant 'Google' decided to use the technology for data analytics using Chain.link which is a decentralized oracle connecting blockchain to off-blockchain data (Castillo et al., 2020; Chain.link, 2020). The company decided to launch the tools to provide a search engine for all cryptocurrencies (Castillo et al., 2020). An example like this clearly shows the use of technology in practical application within big organisations; small companies can profit from the technology the same way but on a smaller scale. Here are the existing implementations:

i. Asset Management

Banco Santander is the first company that launched blockchain-based money transfers known as Santander One Pay FX (Zaballos, 2019). Bank clients benefited from the technology by using a cheaper overseas transfer by reducing expenses on bank manual labour (Business Insider, 2020). Cross border transactions can be very difficult and expensive as the money must go through different banks before it gets authorised and delivered to the recipient. During this process, all the middlemen take fees from the transfer, leaving the small sum at the end for the recipient (Rosic, 2018). Blockchain can eliminate this issue by creating a platform that uses cryptocurrencies where two parties are only getting charged a very small amount in comparison to traditional bank transfers. This would reduce risk management costs which can be crushing for small businesses (Shields, 2019). Assets can also be managed by creating a network where they can be verified for their authenticity, which has already been done by two French entrepreneurs who launched A Watch Certificate service which focuses on authorising second-hand luxury watches through their system to identify fakes. Every customer is equipped with a QR code, linked to the blockchain that allows them to view the results of 42 authenticity checkpoints. It can also be used to resell the authentic watch in the future. These types of services fight against fake replicas which are illegally

sold for full prices of which customers are unaware (Barber, 2020). This service could also be used by small companies for other valuable assets such as cars, art painting etc. Asset tokens stored on the blockchain would increase liquidity and decrease resell risks.

Going one step further, digital asset management has been used by the company Sorare for selling Ethereum based, (NFTs) Non-Fungible Tokens (Mahtab, 2021). Companies as such use the technology to sell digital ownership of media, gifs, images, sounds or videos. Everyone has the access to the content on the services such as Google or Facebook, but the person who purchases the NFT has the rights to the ownership (Clark, 2021). Beeple, 'Everyday – The First 5000 Days' is a digital blockchain artwork worth 69 million dollars as it sold at Christie's auction in 2021, making it the most expensive NFT purchase (Cascone, 2020). From the analysis by Maayan (2019) the main issue which came up from blockchain asset management was the lack of experience in handling such a high volume of data. Also, Birkin, EMEA Wealth & Asset Management Leader at Ernst & Young mentioned technology limitations in mechanism to provide trust (mining) and regulatory uncertainty for real transaction volumes, system interoperability and penetration attempts (Maayan, 2019; Birkin, 2018)

ii. Supply Chain

Another great example where blockchain can be implemented is supply chain management. Real-time tracking of goods throughout the supply chain could benefit the companies as they would know the exact location of their products, minimising the risk of confusion in logistics and loss of goods (Harting et al., 2020). A study by Deloitte and MHI from 2016 shows that 44% of leading supply chain companies find sensors in the products useful and 87% of them plan to use the technology by 2020 (Rosic, 2018). Judging by the statistics, it can be said that companies introduce sensor technology to their supply chain as it helps them to track all of the processes, making the process transparent and more efficient. IBM using QR code tracked 18 million transactions for 17 thousand different products since 2019. IBM's Hyperledger Fabric has been used to operate a food supply chain for IBM Food Trust making the operations easier and transparent simplifying difficult processes (Castillo et al., 2020). Swiss company Nestle also decided to benefit from using IBM Food Trust for their organic and baby foods allowing the customers to scan the QR Code and see the food journeys (Castillo et al., 2020). SyncFab uses QR tokens to provide tracking solutions for aerospace and automotive. The history of an individual item is crucial for the manufacturers as it provides product quality and customer safety (Syncfab, 2020). These

advantages can help small companies by creating a supply chain system that focuses on the smooth management of goods. Small companies could use it internationally for sourcing materials and outsourcing manufacturing to cheaper labour countries (Harting et al., 2020). However, blockchain in the supply chain hides a few inconveniences. The data which goes onto the blockchain is still done manually by an employee who can make a mistake. In the standard supply chain, these problems can be fixed by updating the transaction manually. It is impossible on blockchain and an employee has to enter the data right the first time as it will stay there forever. This could increase the employee's attention however still would not prevent mistakes (Turpitka, 2020). Another important issue that needs addressing is that many big companies will not want to switch their supply chain to the blockchain as it could be very costly which could be a drawback for small businesses as they might lack companies to do business with. It might only be possible until technology grows and becomes cost-effective for bigger companies (Turpitka, 2020).

iii. Real Estate

Individuals could buy, sell, or trade their properties across the countries/continents in an easier manner than it is done today (Morkunas et al, 2019). Creating a blockchain platform on which brokers can easily make transactions of ownership, would make the properties more liquifiable and accessible by

reducing the time it takes to sign all agreements and process legal documents (Rucker, 2018). Authorisation and verification process completed before a user can get on the market would speed up the process, eliminate any complications and reduce the risk of fraud thanks to encryption, offering transparency of the process of the transaction (Business Insider, 2020). Property money would be sent via smart contracts created on the blockchain on keys handover to the new property owner. A property dealer would be present to finalise the deal and make sure all of the steps are followed correctly, eliminating the risk of any fraud which could have appeared by transferring money for the property to the dealer without mutual trust (Rucker, 2018). Unfortunately, This could only be possible if blockchain is adapted and authorised by several companies to make clean and smooth transactions. This would be also much easier if all parties used cryptocurrency for purchases as it would make the transactions more efficient (Rucker, 2018).

iv. Insurance Policy

Transparency and a clear vision of the insurance policies make blockchain technology a great alternative to the current system which insurance companies are using (Business Insider, 2020). InsurePal is a great example of a global provider of decentralized car insurance platform. The blockchain platform provides insurers and policyholders with reduced risk of fraud and enables the fair

pricing of premiums. Using Ethereum public blockchain, the company's solution to overpayment is risk level analysis for the customers through the use of social credit score and The Social Proof TrustScore (Sun et al., 2020). Encrypted information such as driver's history, personal details, health condition and additional information needed to take car insurance are securely stored on the blockchain. All of this information is used to assess the most suitable policy for each customer while also allowing both parties to negotiate the price. It provides utility tokens which are awarded to motivate the participants and build a credit history in a peer-to-peer manner. This is used for future insurance premiums (Sun et al., 2020). Technology can be also used on insured flight tickets when a flight is cancelled or delayed accessing verified flight data sources (Gregorio, 2017). Those benefits could save small companies a lot of money through the legal aspects and preventing expensive insurance policies. Using already existing insurance platforms like InsurePal would be an easier and more affordable option for small businesses unless they are planning to become the next big insurance platform. Nevertheless, the Institute and Faculty of Actuaries released a scientific paper on blockchain in insurance use cases in 2020 and provided several issues with the technology. Decentralization and data leakage are the main reasons halting the introduction of blockchain to insurance policies. Customers

have no authority over their personal information and determining who is eligible to view it, causing legal and security issues for the users (Popovic et al., 2020).

b. Service Analysis

Amazon and Microsoft created blockchain developer services that help investors and businesses in the technical development of their new blockchain solutions. Amazon, using Amazon Web Services (AWS), introduced Amazon Managed Blockchain (Amazon, 2020). It is a network that uses an open-source Hyperledger Fabric framework. It is a scalable and trusted private network which could be great for small businesses due to its advanced privacy control of the network and transaction tracking features which Hyperledger Fabric provides (McClimans, 2019). IBM's website provides a lot of support for the developers (IBM.com, 2020).

Microsoft also developed Azure Blockchain Service which provides the development of smart contracts and built-in consortium management (Zheng et al., 2019). Microsoft states that developers can focus on application development rather than blockchain infrastructure as the system supports multiple ledger protocols. By creating a blockchain network ecosystem, developers can easily eliminate the complexity of the underlying architecture which would be a great option for small businesses as they would not have to start from the beginning (Microsoft, 2020).

c. Literature Review Conclusion

By summarising the literature review and analysing blockchain technology from the technical and business perspective it can be concluded that all of the analysed blockchain solutions could benefit small companies. Yet, some are more advanced due to provided advantages and fewer limitations. The benefit of all-year-round network access and reliability could easily benefit any size company. Using blockchain in the real estate or insurance sector could be beneficial for some small companies. However, blockchain may be great to terminate the need for middlemen in the supply chain, making the system more efficient. The supply chain is also a better solution in comparison to real estate, asset management and insurance policies because it impacts a wider range of small businesses. Real estate could only benefit the real estate industry, the same goes for insurance. Technically asset management could work with a majority of small businesses that deal with the governance of data, but the supply chain can benefit the most from the blockchain.

Manufacturers could buy directly from the suppliers and make smart delivery contracts with transport companies. Transport costs would be reduced by the pre-journey organisation as companies would have a better understanding of the system and logistics. This would minimise unplanned costs due to misunderstanding the routes and

planned stops along the supply chain (Polkowski et al., 2018). Transported goods would be assigned with QR codes and determine a time the transport was ordered, received, and its current location. Visibility of the goods would reduce the friction in the business process (Stanton et al., 2020). This information would stay available and consistent due to the nature of blockchain and could be developed so all parties have an insight into the location of the goods but restrict access to other information. For example, a manufacturer could order raw materials from the supplier finalising all legal necessities all on blockchain and both companies have transparency to the contract. Introducing smart contracts would execute contracts variables to finalise the transaction. In the case of supplier delays to the products, a smart contract would deduct the costs from the payment if necessary (Ali Syed et al., 2019). Monitoring of shipment, better management and less physical paperwork would minimise the place for errors and shorten the delivery time (Polkowski et al., 2018). Smart contracts are a much better solution to traditional physical paperwork because they can be programmable to fit a wide range of documents. Automatic reordering of goods would be processed smoothly by executing previous paperwork through smart contracts. Information on orders could be pre-programmed to make the process faster by producing barcodes,

equipment lists, shipping notice and detailed tracking with pre-defined routes. Alerts could be programmed for when transport is ready to be picked up from the manufacturer (Stanton et al., 2020, p.104). This allows for no human error such as late scheduling of drivers to collect the order and communication issues would be solved by an automatic communication system directly on the blockchain.

Yet, there are some limitations to the use of the blockchain in the supply chain. This could be very difficult to implement new and it might not be cost-effective if only one company is financing the project. To develop a supply chain using blockchain technology, a company requires skilled software engineers with previous blockchain development experience. Software engineer's salary starts from £40,000 a year according to Reed.co.uk (Cheary, 2021). Previously analysed services: Amazon AWS, Microsoft Azure or IBM Hyperledger which are widely available and great choices for the platform to develop the supply chain solution (Nagpal, 2017). Amazon has already been experimenting with the supply chain, allowing users and businesses to send, track and manage goods on the blockchain to see the process. This means that the fundamentals for the supply chain are already in place so the company would only have to modify them for their use, saving time and money on development (McClimans, 2019). Platforms support system monitoring of

CPU, memory, and storage usage but also built-in consortium management using PowerShell and a REST API (Microsoft, 2020). Nevertheless, this project will focus on developing a blockchain web application to demonstrate the fundamentals of how the supply chain can be implemented and outline how difficult it would be to be developed by small companies.

IV. Blockchain Development

a. Software Development Plan

1. Requirements

1.1. Introduction

1.1.1. Purpose

The goal of this software development is to demonstrate and test how easy it would be to develop a supply chain management system on blockchain technology. This will show the difficulties that small companies would have to go through in deciding to employ the technology.

1.1.2. Scope

The main project goal is to represent a blockchain application platform where users such as supplier, manufacturer, distributor, vendor, and a customer can operate in buy-and-sell transactions. The user will create an account using an email address, password and provide an e-wallet number that would be used to purchase and receive cryptocurrency from the sales. User type 'buyer' will be able to buy items from the items list and user type 'seller' will be capable of releasing and

updating items on to the platform. Invoices would be created and stored securely on the blockchain. All users will be capable of viewing invoices corresponding to their accounts. The rest of the system requirements are provided in Appendix A, divided into functional and non-functional requirements.

This project will only focus on demonstrating proof of concept for supply chain management blockchain application and will not be a fully functional application that could be used in a business scenario. It will demonstrate the viability and level of difficulty for small companies to develop blockchain applications.

1.1.3. Overview

For development methodology, Waterfall has been considered due to its wide use in the industry. Unfortunately, it focuses on sequential development by relying on previous tasks. It is also more suitable for large projects (Ahmed, 2012). This project is small in reference to the "IS/IT project size classification methodology" table by Ozteran et al., 2019 as it is only a proof of concept rather than fully functional ready for deployment software (Ozteran et al., 2019). It is a reason why the Waterfall methodology will not work well in this project. Agile would have been a much better option for this project as it is a small project orientated methodology, which focuses on working code rather than writing documentations (Dooley, 2011). However, this project will not use traditional

agile methodology, but the Test-Driven Development (TDD). It shows more potential to be successful in comparison to the previous two methodologies due to constant testing throughout the project, rather than development and testing at the end (Ahmed, 2012). Test cases will be written throughout the development in black box testing (Appendix I) which are elaborated in section IV-a-2.5. A study involving Microsoft and IBM shows 15-35% faster development using TDD (Aston, 2020). The project will be conducted by one software developer which means only one unit can be developed at a time. For this, TDD will be the best methodology approach, because software requirements also test cases that will be used to test if the developed unit has met project requirements. This minimises the time on software documentation which gives more time to research and develop the software. The strategy of unit testing will be used with a dynamic bottom-up programming style that focuses on the development of specific units combining them at the end to achieve the fully functional software (Rauss & Pourtois, 2013). The software will be tested along the way, noticeable and important issues will be documented in the Issue Log (Appendix J). This will be used in case of switching dynamically between different test cases to prevent losing track of the progress and unsolved issues in the design (Smartbear.com, 2021). In choosing

a programming language, assessing which programming language fit the criteria, purpose, and scope of this project was important, as using the TDD methodology, the chosen language needs to be adaptable. Python is one of the most versatile programming languages with many plugins and libraries for back-end web application development. In comparison to Java and C++, Python has the advantage of shorter development time as it takes less code to achieve the same results (Python, 2021). It is the second most popular programming language and has already been used to develop Ethereum and Hyperledger Fabrics, therefore it will be the best programming language for this project (Chang, 2020).

For the front end of the web application needs to be consistent in design, HTML and CSS will be used as they provide tools to present the blockchain application creating a link between the code and visualisations necessary for the user to interact with the system. HTML is a highly effective tool in achieving this because it provides tools for visualization which can be accessed through the web browser. The web application will also contain JavaScript which will be used for web scripting. JavaScript does not require access to memory and CPU making it a good choice for easy and simple web scripting which will be used in this project (JavaScript.info, 2021).

Meetings of 30min with the project supervisor will be scheduled

every week to report the project accomplishments. The whole project will be monitored and managed using Gantt Chart which can be found on Google Drive:

(hidden)

Software version control will be managed with the GitHub repository:

https://github.com/xMPon/Git_Computing_Project.git

1.1. Risk Assessment

Risk Management Table (Appendix B) and Risk Matrix (Appendix C) has been designed using likelihood and impact assessment. This was done to control the risks and find the most likely to cause issues during the project. It was important to find them before the project started to analyse mitigation actions before they could occur (Luckey & Phillips, 2006). Risk Matrix shows that risks 2,3,4 and 6 seem to be the biggest threats for the project. Solutions to the problems and mitigation actions can be found in Appendix B.

Risk 3 has the higher probability and impact ratio which classifies as the most dangerous of the risks listed. Blockchain development can be very advanced in itself, combining it with supply chain web application can take a very long time to develop which means some of the features will not be possible to develop in time. Following mitigation actions minimises the risk likelihood. Risks 2, 4, 6 has also received a high impact to likelihood ratio which makes them risks that have to be considered and

acted on before they cause big issues in the development process.

2. Design

2.1. Use Case Models

Use case models can be found in appendix D showing the different type of users interacting with the system. Node 1 and 2 can also be a buyer or a seller, they are independent of the transactions made between different sellers and buyers for decentralization and security.

2.2. Architectural Design

Architecture design can be found in appendix E. It represents all system attributes and the connection between them. Each one of the users has access to the system through connected to the internet mobile device. Requests for transactions such as purchases, and invoices are created by the web-based blockchain application and authenticated through smart contracts. After creating a valid block by one of the clients (miners), this transaction data is permanently added to the blockchain. A blockchain system is a peer-to-peer network which means that all the data has to go through different clients. Technically this is done outside of the internet on the physical hardware machines however, to represent that system is carried out somewhere else it has been placed inside the Internet cloud.

2.3. Software Design

The Unified Modeling Language diagram in appendix F visualizes the system design by

dividing software architecture into components and subcomponents which are going to be used as a blueprint for software development. Appendix D, E and F will be also used in the development process as supporting materials in understanding the complexity of blockchain development.

2.4. Interface Design

The graphical user interface of the web application will have a simplistic design which is demonstrated in Appendix G. The background will stay black to reduce eye strain in low-light conditions. The colour of the text will be white and green to maximise the contrast with the background, also helping to reduce eye strain. The main design colours and design specifications can be found in Appendix H. The user interface will have a navigation bar which will be used to switch between the screens with a consistent layout. The navigation bar will be presented at the top of the screen. Some screen resizing optimisations will be introduced to help the user interact with the web application on different devices. Logout buttons will exist on each of the web pages for easy exit. Depending on the user type there will be different functions for the user to interact with. For example, the 'seller' user type will have the opportunity to create and update items that will be submitted on the blockchain for other users to see and purchase. The 'buyer' user type will be able to choose a quantity of the items to buy, using buttons and textboxes etc. The

design will follow design standards such as button cursor, checkboxes and input fields which are used across the words to minimise confusion from the design. Submit buttons will be placed next to a field with which they are interacting with. The simple choice of design was influenced by methodology as this project will function over aesthetics.

2.5. Test Plan

The software will be tested by the black box method shown in Appendix I followed by GUI designs in Appendix H to confirm that all functional and non-functional requirements described and analysed in the project planning phase (also Appendix A), have been met to classify software development as successful. Methods such as Acceptance, Performance or Compatibility Testing, have been taken into consideration. However, black-box testing proved to be the best option for this project, because TDD methodology is based on creating requirements which are also test cases. The requirements will be created throughout the development in the Appendix A and elaborated on in Appendix I simultaneously, which allows to test individual unit requirements throughout the development process. (Smartbear.com, 2021). This method will show which units have not been yet finished, improve software quality, and mitigates the risk of software failure (ReQtest.com, 2021). The GUI testing will be based on the design from Appendix H and meeting functional

requirements from Appendix I to make sure all features such as buttons, text fields etc. are usable. At least two users will be asked to test the application to receive constructive feedback on usability, navigation, and functionality.

b. Software Testing

2.6. Functional Testing

Several issues have not been dealt with in the development process and need to be declared. The first major issue is that dynamic files updated on the blockchain have not been fully tested and the lists storing the data on users, invoices and items are not actively applied to the blockchain's blocks. Blocks are also created when the hash number is calculated, but the mining of the block is not corresponding to the system of real blockchain, because there are no miners who receive a reward for the process. However, a user who runs the programme is also triggering the mining mechanism so that blocks are created. Blockchain is encrypted with SHA256 as it was initially planned. However, the front end of the web application has no encryption, and the lists are saved in plain text, creating a security risk. This is easily fixable and could be improved in the future updates. The session timer has been set to 10 minutes and will log the user out once the time is up if inactive. Further test analysis can be found in Test Plan (Appendix I).

2.7. Non-Functional Testing

Testing shows that 7/10 (70%) of the non-functional requirements shown in Test Plan (Appendix I) has been successful with exceptions on security features of manipulating the data outside of the designated section (QR1.2.2.8) and confidential data should not be accessible by unauthorised users (QR1.2.2.10). This can be improved in the feature updates however it does not raise a concern in the proof of the concept. Web navigation does not raise any concerns; if the system is up and running, the user can log in and make the account at any given time without a problem.

2.8. GUI Testing

Functional testing has been carried out to make sure other users can interact with the system without a problem. Web application features such as radio buttons, text boxes and buttons follow correct HCI fundamentals by displaying and highlighting the fields with the cursor. The navigation features are consistent and stay at the same place for easy web page movement while the colouristic are easy to read. Two independent users have tested the application and declared that all of the text and buttons are easily understandable and are not difficult to remember. The overall design is pleasant to look at and meets the software criteria from section IV-a-2.4 and Appendix H. Nevertheless, the user's feedback concluded that the feature missing from the development process is error messages which gives the

user feedback on their actions. The input fields are functional but only if the user knows how to operate with them in the first place. What has also been noticed is the lack of adaptation to different screens which should have been improved as the text can overlap making the content unreadable.

c. Results

The applied methodology has been very successful due to unit testing during the development process. A simple code has been produced to fail in order to be built upon afterwards to meet software requirements by consistent testing and the improvement cycle. Dynamic bottom-up programming has been very efficient as it focused on the development of individual units which has been broken down into smaller tasks. Blockchain architecture has been developed at first to see the complexity of the technology as it was also the main scope of this project, followed by the development of front-end using HTML, CSS and some dynamic website feature using JavaScript. Due to the vast popularity of the Python programming language online, it has been relatively easy in finding supporting and relevant materials which helped in the development process. One of the main sources to mention is StackOverflow.com. An online forum for software developers and technologists (StackOverflow.com, 2021). This has been only used to find architectural solutions for issues which came up in Appendix J. It has supporting materials for

Python but also CSS, HTML and JavaScript used in this project.

The supply chain web application is functional on the local machine and meets the majority of the requirements (from Appendix A) which is shown in the Test Plan (Appendix I). Users after registration can log in to the system, choose the user type and release and purchase items. The transactions are displayed in the form of invoices which users can see and are permanently stored on the blockchain. The project also covers the fundamentals aspects of blockchain technology. The blocks are created using POW mining when the blockchain starts.

Development focused on functionality rather than graphical design and so the simple version of GUI has been successfully produced. The interface has been developed according to the GUI design from Appendix H and tested by two users. There have been some inconveniences in the design such as no feedback on user's actions but overall, the design has passed the test.

Due to good time management, Gantt Chart and Risk Analysis (Appendix B), there have not been many issues causing trouble in the software development process. The only risks which have to be addressed are Risk 2 (Scope for the project is too ambiguous) and Risk 3 (Software development takes longer than initially planned) from Risk Matrix (Appendix C). The developed software has many places for improvement starting with validation and

mining to the front-end graphical interface. The time frame for the project has limited what can be developed to necessary features and functions which has been organised and tacked on from the start of project management. It is the reason why those risks did not cause trouble and have been dealt with straightforwardly.

Test Plan (Appendix I) shows few issues which could be improved on in future updates, but it is not crucial for this proof of concept. The mining should be improved, it is based on Bitcoin mining algorithms, but it is extremely inefficient and does not involve cryptocurrency. Without the cryptocurrency, independent miners would not be interested in the mining process as it will not bring them any value. The e-wallet has not been developed to send and receive cryptocurrency for transactions. The users 'miners' are also not created to store a full copy of the blockchain and mine new blocks. Instead, those functions are done locally on the user's machine. Another issue comes with user 'seller' updating already existing items. The system only allows the user to update the first instance in the list and not the other once. The problem also comes from the dynamic update on the information displayed. The system displays some information only after the user logs out and log back into the system. All of these issues can also be found in the Issue Log (Appendix J) for further reference.

Documentation such as Version control using GitHub has been very useful due to easy access to the code while working on different machines. It has also created the security feature as the data was kept organised on the cloud rather than the local machine. Gantt Chart has also been a great management tool as it mobilised and provided the time frame for the development process. Gantt Chart and sprint backlog with project supervisor can be found in section IV-a-1.1.3. The sprints have been organised for reference on further steps from the meetings with the project supervisor. They have been very helpful in the software and project development process.

V. Final Words

a. Summary

Blockchain can be applied by all size businesses as it was analysed in this project. Corporations such as IBM and Walmart to name a few have already developed solutions that help their business in logistics, supply chain and asset management (Dimitrov, 2019). Medium-size businesses such as InsurePal are using blockchain for car insurance policies and it will not take long until companies start adapting the technology to Real Estate management and auction services (Sun et al., 2020). Small businesses can also benefit as the analysis proves by creating a new blockchain with cryptocurrencies or even using already existing solutions from Google, Amazon, or

IBM (Nagpal, 2017). Small companies could create the blockchain themselves but using already existing solutions would be a better decision. The project has outlined the benefits which come from blockchain technology but also ways that help the businesses in non-profitable aspects which make the companies more efficient, secure, and reliable. There are limitations such as governance which small companies would have to overcome nevertheless, companies have the options to choose from and as the technology is still growing it will only reveal new benefits and solutions to the limitations (Business Insider, 2020; Castillo et al., 2020).

b. Implication

By analysing the technology and concluding the best practices for the blockchain, small companies could use it in their decision process by selecting from the results which application would be most beneficial for their business. Research has covered the vast majority and type of businesses as it focused on helping small businesses rather than focusing on a specific sector like finance or HR. However, businesses in retail or manufacturing could use the blockchain for supply chain management or logistics. Companies that are based in real estate could use the technology for house auctions or asset management. All small companies could benefit from the insurance policies on the blockchain and financial services such as cryptocurrencies and overseas money

transfers. This just shows the scale on which the technology can be used.

c. Discussion

This project focused on the blockchain exclusively, but it does not mean that value cannot be created in conjunction with other emerging technologies to provide small companies with better opportunities. There has been a lot of analysis from other researchers of how blockchain and the internet of things could work together to improve the traceability of the goods in the supply chain (Feng et al., 2020; Novo, 2018). The benefits are much greater while using both of the technologies combined as they complement each other very well giving the businesses and customers inside the life-tracking of the goods. This would be most beneficial for companies which deal with fresh or life products with an expiry date. Tracking those types of products would guarantee freshness and high-quality standards (Feng et al., 2020).

There are approximately 7,130 micro, small, and medium businesses in the food and drink industry according to National Statistics (2019) which could benefit right now from the invention (GOV.uk, 2020). The biggest multinational retailer, Walmart has already implemented blockchain in their supply chains, it is time for small businesses to take the same opportunity and use it before the competition see the advantages first and sweep them out of the market (Dimitrov, 2019). Research shows that blockchain technology can be used

for its financial benefits by cutting the cost on middlemen like banks or small security expenses. Nevertheless, blockchain does not necessarily generate profit or save companies on huge expenses but in the long run, it creates market value by creating a stable and secure platform that companies can use to ensure their products quality, authenticity, or uniqueness. The security that blockchain provides can be used as a marketing strategy for the company's advertisement. The supply chain was one of the better applications for the blockchain, but it does not mean it is the only use. The analysis showed that blockchain can also be used in different sectors for processing payments, storing goods, cross-industry integration, and enhanced user authorizations (Turpitka, 2020). Even so, it is a great technology that is constantly evolving, and it is a matter of time when someone is going to find the ultimate solution which all other companies will follow, like it was done by Apple releasing Macintosh, 3 years after IBM's personal computer was presented in 1981 (Frangoul, 2018). To see the results, look at IBM's and Apple's brand valuation.

d. Conclusion

This project aimed to find benefits of blockchain technology for small businesses. Overall, the project has proven that blockchain can be used by many small companies due to the vast coverage of different business fields, from the supply chain to asset management. After all, the supply chain ended up being one

of the best uses for the blockchain technology during this project, as it shows the most potential and usefulness. Many small companies which deal with physical products, transport of goods etc. would benefit from the benefits such as reliability, security, and consistency (Harting et al., 2020). However, there are limitations which those companies would have to face. The developed proof of concept has underlined that building blockchain from the ground up can be challenging in the real-world scenario and small companies might not have the resources to deal with them. Limitations such as governance or block mining are just two examples. One of the solutions presented in this project was the Hyperledger Fabric which is an already existing solution, great for small companies to start their journey with blockchain, as they will not have to produce the technology from the start, worrying about the mining process or architecture of the blockchain (Yang, 2019).

Small companies can use the technology and revolutionise sectors. Supply chain management is the best suited one, however asset management has shown potential too. The invention of NFT sales has been very profitable for small companies such as Sorare and the market could expand in upcoming years (Mahtab, 2021).

When it comes to the software development process in this project, it cannot be confirmed that test-driven methodology

works on larger projects as it was not tested nor was it the purpose of this project. On the other hand, from the results of software development, it can be confirmed that TDD development with bottom-up programming is effective and can be used in any small software project where there is only one developer. The black-box testing has been efficient in dynamic unit testing and most of the issues have been solved in a relatively short time. The overall development process has demonstrated the proof of concept for supply chain application based on blockchain technology which could be used by small companies to establish the complexity of blockchain architecture and fundamentals that are required to create a similar application.

e. Recommendations

Recommendations from the analysis for small companies are to focus on already existing blockchains such as Hyperledger Fabric which are designed for enterprises in mind as most of the fundamental structure is already provided (Yang, 2019). Businesses could build supply chain managements based on Hyperledger Fabric, which has great internet support on their website and online forums. Using already existing platforms such as Amazon AWS or Microsoft Azure would cost the companies less, as software engineers would spend less of their time producing the solutions.

Prototype

created during this project has been a very simple adaptation of the real supply chain management which would not work in the real-life scenario. The summary and discussion section have outlined which features could and should be implemented to make the application usable in a real business case. The prototype has shown the difficulty of the technology and how much time and money it would have to be spent for the investment to work and become profitable. Therefore, it would be much better for smaller companies to start with the development of an already existing blockchain and use provided support to adapt it for their business needs.

Choosing the blockchain solution would have to depend on the company and their sector. A supply chain can be adapted by companies that rely on the supply and demand of goods such as manufacturers, retailers etc. but any of the analysed solutions would be a great alternative to the current system for small businesses when overcoming the limitations.

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Appendix

a) Appendix A System Requirements

1.2.1.	Functional Requirements
FR1.2.1.1.	Accessible web application through a browser search

FR1.2.1.2.	Ability to register using an email address, password and e-wallet code
FR1.2.1.3.	An email has to be valid
FR1.2.1.4.	Email address cannot be used twice to create the account
FR1.2.1.5.	Password is a combination of letters, number and symbols
FR1.2.1.6.	The second registration screen allows user to choose between two types of the accounts
FR1.2.1.7.	Ability to login using previously registered accreditation
FR1.2.1.8.	User can only access the account information by providing registered accreditations existing on the blockchain
FR1.2.1.9.	Web page displays account details
FR1.2.1.10.	User can only view their profile accreditations
FR1.2.1.11.	Ability to view the connected e-wallet details
FR1.2.1.12.	The web page displays the user's log of previous invoices
FR1.2.1.13.	User can only view his past invoices
FR1.2.1.14.	Ability to navigate through different web pages
FR1.2.1.15.	Ability as a seller to add items to the market
FR1.2.1.16.	Ability to update the item quantity
FR1.2.1.17.	Ability to update the item price
FR1.2.1.18.	Ability to update the item description
FR1.2.1.19.	Ability as a buyer to purchase items from the market
FR1.2.1.20.	Ability to log out from the account
FR1.2.1.21.	Automatic system creating new blocks by mining
FR1.2.1.22.	Transactions are encrypted with SHA256
FR1.2.1.23.	Transactions are referenced with an encrypted timestamp

FR1.2.1.24.	Transactions are stored on the encrypted blockchain
1.2.2.	Non-Functional Requirements
QR1.2.2.1.	User can register at any given time
QR1.2.2.2.	User can easily create the account
QR1.2.2.3.	The web application is available 24/7
QR1.2.2.4.	The web application can be easily understood and navigated
QR1.2.2.5.	The web application can be easily navigated
QR1.2.2.6.	User information is updated immediately
QR1.2.2.7.	Only provided necessary information is kept on the blockchain
QR1.2.2.8.	Blockchain data cannot be manipulated or changed outside of the dedicated sections
QR1.2.2.9.	Errors should be redirected to the corresponding web page
QR1.2.2.10.	Confidential data should not be accessible for the public or unauthorised users

b) Appendix B Risk Assessment

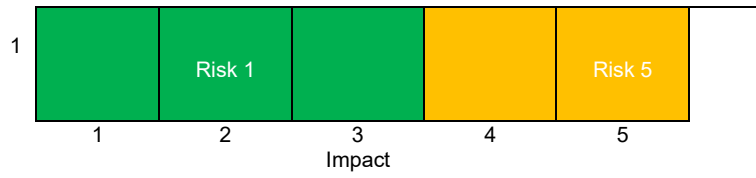
	Risk Category	Risk Description	Likelihood	Impact	Mitigation Actions	Contingent Actions
1	Technical	A new version of the software does not support developed software	Rare (1)	Minor (2)	Update the code to GitHub for the version control	check the update and the features which have been changed to see where the problem is
2	Technical	Scope for the project is too ambiguous	Likely (4)	Moderate (3)	Analyse which features and functions can be developed in the time frame considering the Gantt Chart	Reflect on the development plan to make sure all the key components are developed and cut down on less important features

3	Technical	Software development takes longer than initially planned	Likely (4)	Major (4)	Follow the Gantt Chart to fit in the time frame, estimate the length which takes to develop each task	Consider the options and which tasks can be skimmed
4	Technical	Blockchain is unable to store lists of data	Possible (3)	Major (4)	Research how data can be stored on the blockchain and what are the limitations	Consider storing only the most important data on the blockchain to fit the criteria
5	Technical	Hardware failure	Rare (1)	Critical (5)	Keep all of the files backed up on Google drive at all time, code from PyCharm should be systematically backed up on GitHub	If possible, retrieve the data and fix the hardware. While waiting, focus on other parts which can be developed without access to hardware
6	External	Covid-19 does not allow for further development	Possible (3)	Major (4)	Prevent catching Covid-19 by wearing a face mask, washing hands, and following government guidelines	Update the supervisor on the situation immediately and consider taking a grace period to catch up on project development
7	External	Becoming ill and unwell to continue with the project	Unlikely (2)	Major (4)	Eat healthy and nutritious food, dress appropriately to the weather and stay away from infections	Contact the doctor for advice, take prescribed medicine and recover. Update the supervisor on the situation for further assistance

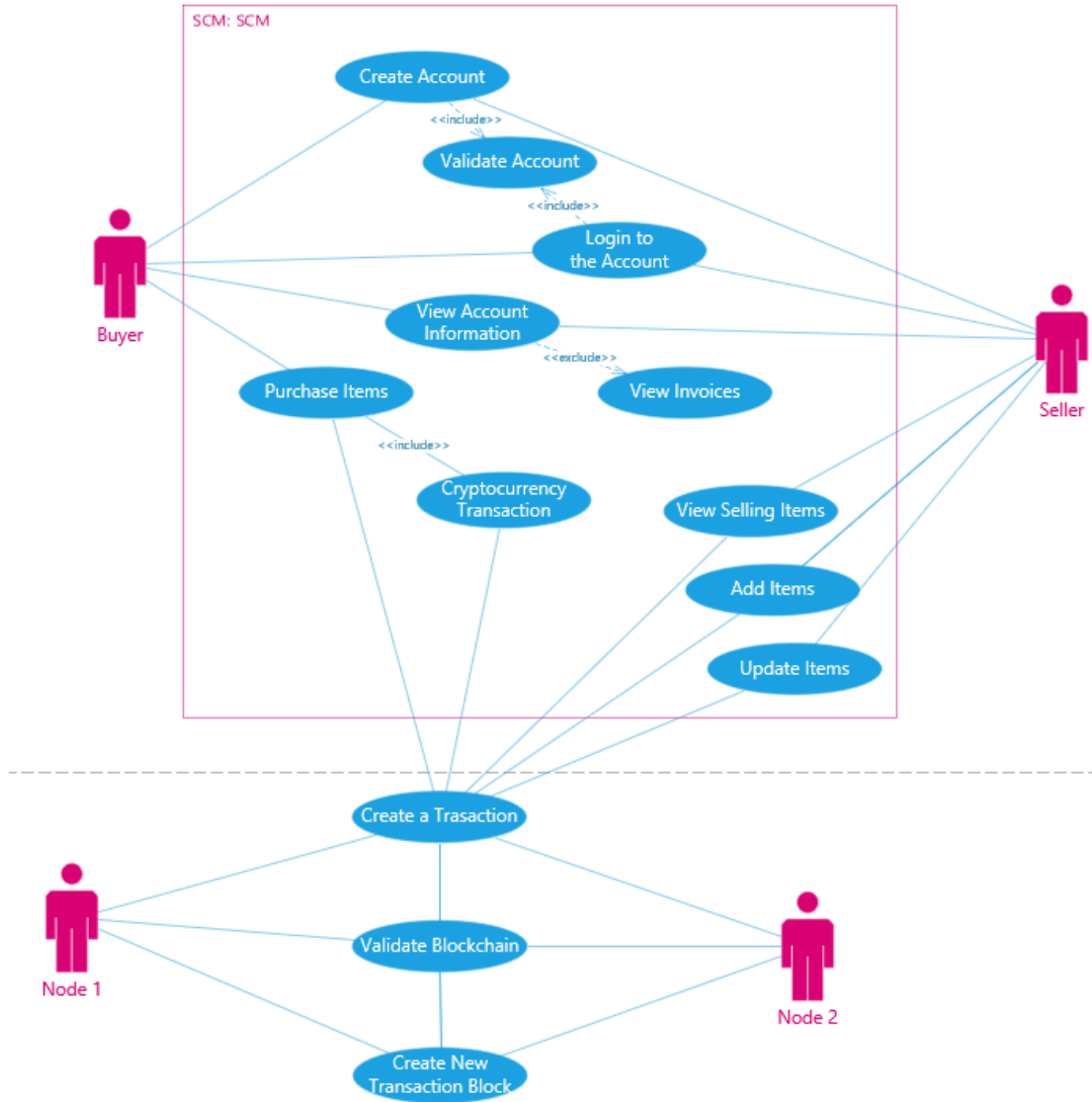
c) Appendix C Risk Matrix

5					
4			Risk 2	Risk 3	
Likelihood 3				Risk 4 Risk 6	
2				Risk 7	

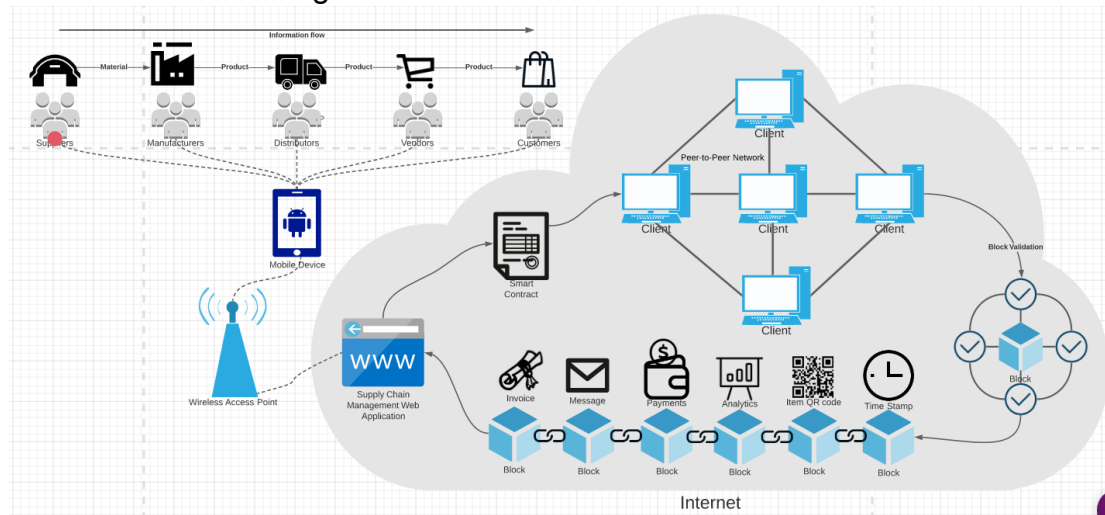
Rating	Likelihood	Impact
1	Rate	Insignificant
2	Unlikely	Minor
3	Possible	Moderate
4	Likely	Major
5	Certain	Critical



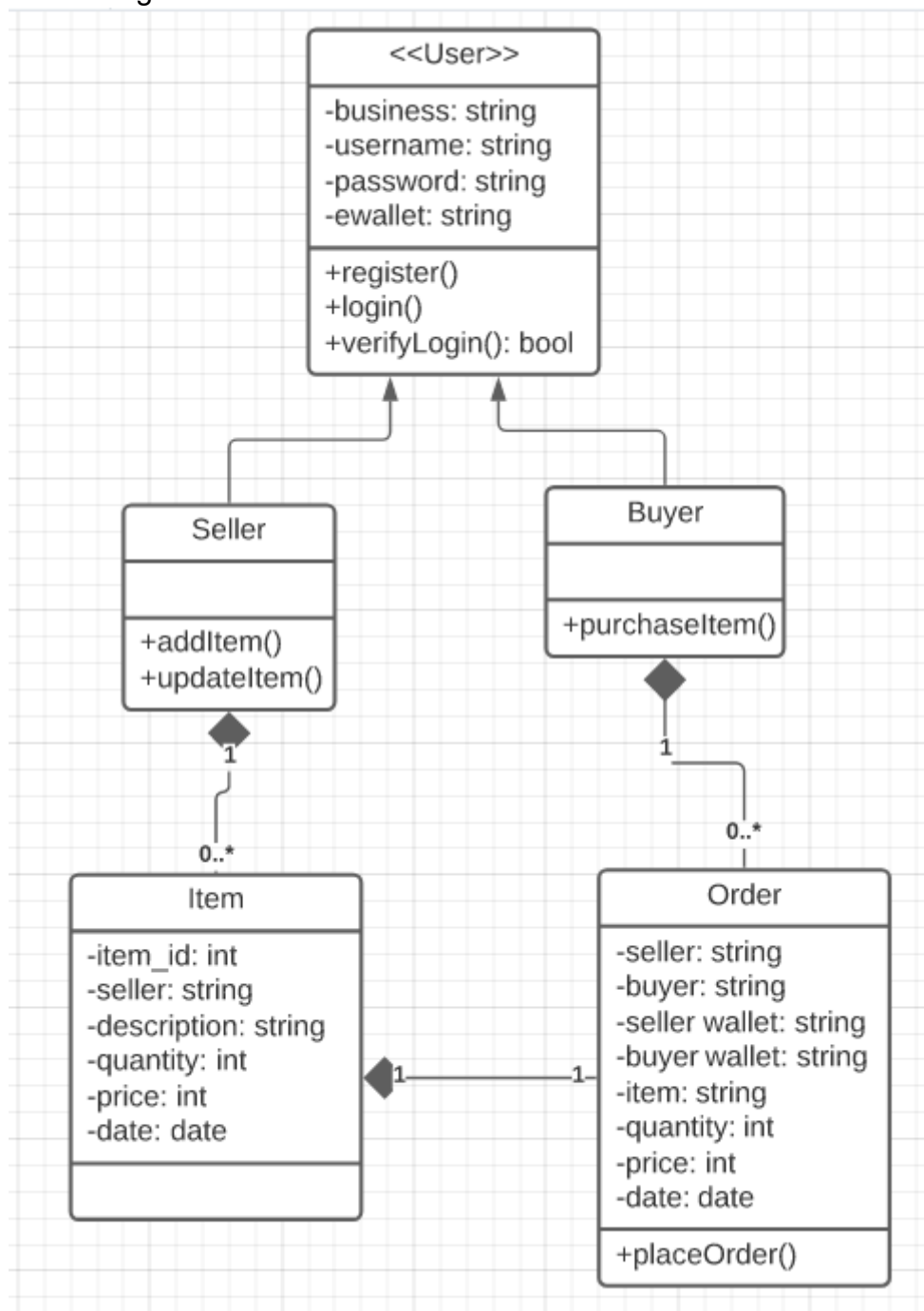
d) Appendix D Use Case Model



e) Appendix E Architecture Diagram



f) Appendix F UML Diagram



g) Appendix G GUI Design

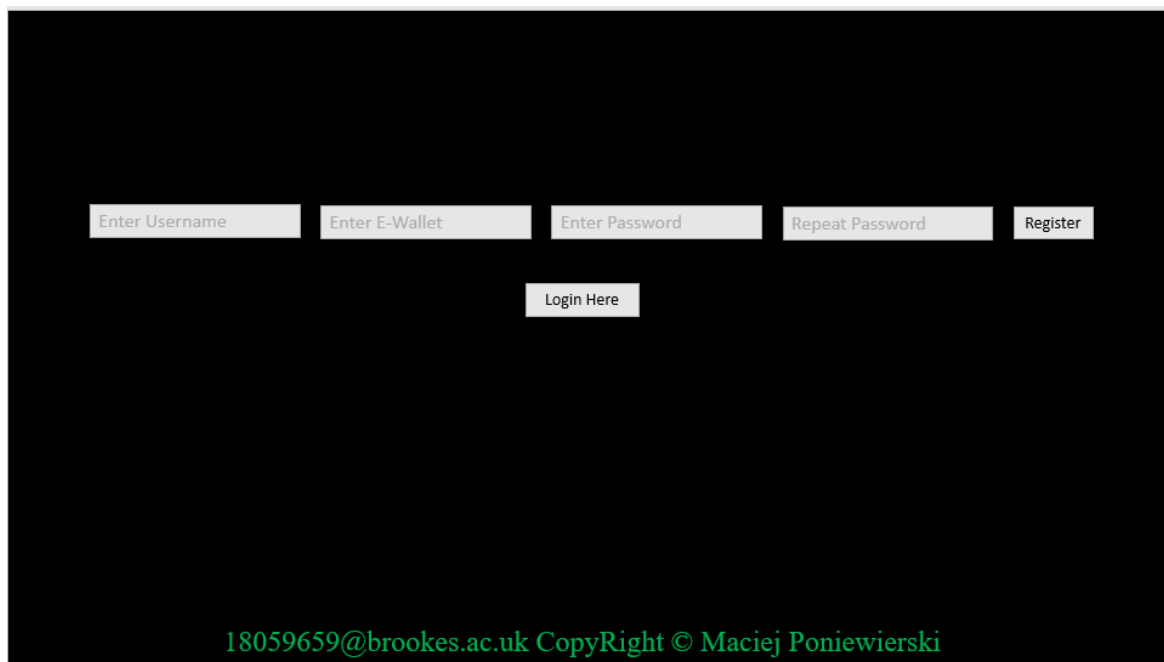


A login interface with a black background. It features three input fields in a row: "Enter Username", "Enter Password", and a "Login" button. Below these is a "Sign Up Here" button. At the bottom, there is a green text string: "18059659@brookes.ac.uk CopyRight © Maciej Poniewierski".

Enter Username Enter Password Login

Sign Up Here

18059659@brookes.ac.uk CopyRight © Maciej Poniewierski



A registration interface with a black background. It features five input fields in a row: "Enter Username", "Enter E-Wallet", "Enter Password", "Repeat Password", and a "Register" button. Below these is a "Login Here" button. At the bottom, there is a green text string: "18059659@brookes.ac.uk CopyRight © Maciej Poniewierski".

Enter Username Enter E-Wallet Enter Password Repeat Password Register

Login Here

18059659@brookes.ac.uk CopyRight © Maciej Poniewierski

Choose One:

buyer

seller

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Add Order

Order History

Account Info

Close X

<user type>

Profile Username: <username>

Account Ewallet: <ewallet>

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[Add Order](#) [Order History](#) [Account Info](#)

Close X

SELLER	BUYER	DESCRIPTION	QUANTITY	PRICE	DATE
<user name>	<user name>	<item description>	<item quantity>	<item price>	<uploaded date>
<user name>	<user name>	<item description>	<item quantity>	<item price>	<uploaded date>
<user name>	<user name>	<item description>	<item quantity>	<item price>	<uploaded date>
<user name>	<user name>	<item description>	<item quantity>	<item price>	<uploaded date>
<user name>	<user name>	<item description>	<item quantity>	<item price>	<uploaded date>

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[Add Order](#) [Order History](#) [Account Info](#)

Close X

Description	Quantity	Price	Added	
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•
<item description>	<quantity>	<price>	<date>	•

Add New Item:

Item Description

Quantity

Price

Add Item

Update Existing Item

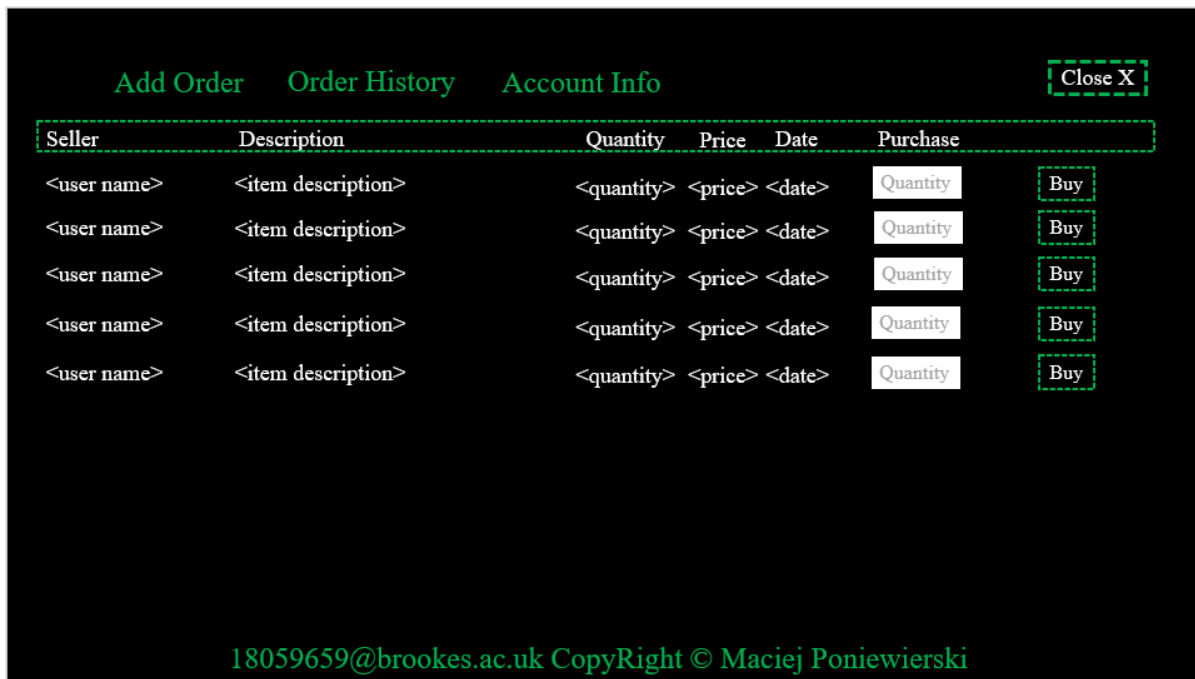
<description>

<quantity>

<price>

Update Item

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h) Appendix H GUI Design Specifications

Design Specifications			
Text	Sans-Serif		
Font			
Design		#000	Background
Colours		#008000	Text
		#FFF	Text
		#444	Text

i) Appendix I Test Plan

Test Number	Where	Action	Expected Outcome	Actual Outcome	Test Result	Test Comments
1	PyCharm Software	Run the programme	Redirect to the start screen without any errors	Web application starts in the browser	✓	
2	Login Page	Start screen	Login page displayed	Web application loads login page	✓	
3	Registration Page	Submit button register button	Registration page displayed	Web application loads registration page	✓	

4	Registration Page	Submit button all entered values	Redirect to the choice page	Web application loads a choice page after correct registration submission		
5	Registration Page	Submit button incorrect email value	Reload the page and ask the user to try to register again	Reloads the page correctly		The system does not support error messages informing the user
6	Registration Page	Submit button different passwords	Reload the page and ask the user to try to register again	Reloads the page correctly		The system does not support error messages informing the user
7	Registration Page	Submit button without one of the values	Reload the page and ask the user to try to register again	Reloads the page correctly		The system does not support error messages informing the user
8	Registration Page	Submit button with no values	Reload the page and ask the user to try to register again	Reloads the page correctly		The system does not support error messages informing the user
9	Choice Page	Submit button type seller	The account is assigned with seller option and redirects to the order page	User type is assigned with 'seller'		
10	Choice Page	Submit button type buyer	The account is assigned with buyer option and redirects to Add order page	User type is assigned with 'buyer'		
11	Login Page	Submit button using registered accreditations	Compare submitted accreditations to the list and if the username and password matches, redirect to Add Order	User can successfully login by providing existing accreditations		
12	Profile Page	Navigate to account info	Display user's accreditation	The profile page displays necessary account information		
13	Add Order Page (as seller)	Navigate to add order	Display functions add item and list of current items in the list	Display the correct information		The list is not dynamically updated when the seller adds a new item
14	Add Order Page (as seller)	Submit button new item	Add the item details to the list and display them on the Add Order Page	The items are added to the list but not displayed		The list is not dynamically updated when the seller adds a new item

15	Add Order Page (as seller)	Choose an item to update with the checkbox function	Display change item function with the already existing information of the item	User can choose the first item to be updated		The system only allows the first item to be updated
16	Add Order Page (as seller)	Submit button update	Changes the item information as followed in the submitted information	Information is only changed for the first item		The system only allows the first item to be updated
17	Add Order Page (as the buyer)	Navigate to Add Order	Display list of all items in the list	A list of all items created by the sellers is displayed with the item description, quantity price etc.		
18	Add Order Page (as the buyer)	Select the item and provide the quantity	The quantity provided for the time is displayed in the designated section	The quantity is added and displayed correctly		
19	Add Order Page (as the buyer)	Submit button purchase	Create an invoice in the list with information on the item, seller, buyer, and quantity, subtract the purchased quantity from the item quantity, update the list (if the quantity is zero then delete the item from the list)	The system creates an invoice and deducts the quantity from the item in the list		The invoice does not appear automatically
20	Order History Page	Navigate to order history	Display all invoices relevant to the account username	Information is displayed correctly by the account username		
21	Error 404 Page	Error 404	Error handler redirects to Error 404 Page	Page goes to Error 404 Page		
22	Error 405 Page	Error 405	Error handler redirects to Error 405 Page	Page goes to Error 405 Page		
23	Error 500 Page	Error 500	Error handler redirects to Error 500 Page	Page goes to Error 500 Page		
24	Background	N/A	All lists are updated to 'data' on the block	Lists are added straight after the second block is created with all the data found in the lists, including user's information, items and invoices		
25	Background	N/A	Blockchain running in the background	Blockchain runs in the background if 'MaChain.py' and 'connector.py' are run separately in two windows		

26	Background	N/A	The programme creates new blocks automatically	The programme runs constantly creating new block when it solves the nonce difficulty		
27	Background	N/A	The blocks are validated each time a new block is created	N/A		Function for validation has been created but it has not been tested appropriately
28	Background	N/A	Ability to view the connected e-wallet details	N/A		The function of an e-wallet has not been created
29	Background	N/A	Transactions are encrypted with SHA256	Encryption works correctly on the blockchain transactions		The front end of the application is not encrypted
30	Background	N/A	Transactions are referenced with an encrypted timestamp	The timestamp is added when a new block is created		
31	Background	N/A	Transactions are stored on the encrypted blockchain	Block is encrypted with SHA256		
32	Non-Functional System Performance	N/A	User can register at any given time	User can register at any given time when the application is running		
33	Non-Functional System Performance	Register new account	User can easily create the account	User can easily create an account when the application is running		
34	Non-Functional System Performance	Interact with the system at different times when the system is running	The web application is available 24/7	The web application is available 24/7 when the application is up and running		Application has not been uploaded to the online system and only works locally but it has never been the requirement
35	Non-Functional System Performance	Interact with the system	The web application can be easily understood and navigated	User can easily navigate through the web pages		
36	Non-Functional System Performance	Interact with the system	The web application can be easily navigated	User can easily navigate through the web pages		

37	Non-Functional System Performance	N/A	User information is updated immediately	Some information is updated immediately		Partially correct as information are dynamically updated to the lists but not always display actively on the web page
38	Non-Functional System Performance	N/A	Only provided necessary information is kept on the blockchain	Only lists of users, items and invoices are kept on the blockchain		
39	Non-Functional System Performance	N/A	Blockchain data cannot be manipulated or changed outside of the dedicated sections	From a user perspective, the data cannot be accessed but experienced hacker might find a way to do it		Partially true as the front end of the application is not encrypted
40	Non-Functional System Performance	N/A	Errors should be redirected to the corresponding web page	Errors are redirected to appropriate web pages (404, 405, 500)		
41	Non-Functional System Performance	N/A	Confidential data should not be accessible for the public or authorised users	From a user perspective, the data cannot be accessed but experienced hacker might find a way to do it		Partially true as the front end of the application is not encrypted

j) Appendix J Issue Log

No.	Issue Description	Reported	Status
1	Block is not created	16/12/2020	Closed
2	New blocks are not created	16/12/2020	Closed
3	New blocks are not added to the list	23/12/2020	Closed
4	Only one block is created after program starts	23/12/2020	Closed
5	Cannot access previous hash number	27/12/2020	Closed
6	Blockchain does not appear	29/12/2020	Closed
7	Block do not append encrypted	05/01/2021	Closed

8	Navigation bar is not working	05/01/2021	Closed
9	User inputs are not redirected	05/01/2021	Closed
10	User cannot login	12/01/2021	Closed
11	Register function is not fining accreditations	16/01/2021	Closed
12	Choice page does not pass information over	17/01/2021	Closed
13	Footer does not display correctly	29/01/2021	Closed
14	Styles are not inherited by the web pages	30/01/2021	Closed
15	User type 'Seller' cannot create items	07/02/2021	Closed
16	User type 'Seller' cannot update all items	19/02/2021	On Hold
17	User type 'Buyer' cannot input quantity	19/02/2021	Closed
18	Logout button does not work	20/02/2021	Closed
19	Session function does not work	24/02/2021	Closed
20	Session crashed before starting	27/02/2021	Closed
21	User is not redirected to the index page	06/03/2021	Closed
22	Navigation bar does not display correctly	06/03/2021	Closed
23	E-wallet number is not used	06/03/2021	On Hold
24	Lists are not compatible with the blockchain	07/03/2021	Closed
25	Screen resizing is not scalable	10/03/2021	On Hold
26	Mining function does not work	10/03/2021	Closed
27	Mining function appears to be in infinite loop	13/03/2021	Closed